

NCC VOLUME TWO, ENERGY EFFICIENCY

Frequently Asked Questions (FAQs)

QUESTION 1

What type of software may be used under the energy efficiency Verification Method?

Answer

The Verification Method, V2.6.2.2, requires the use of a calculation method that can demonstrate that the proposed house is no less energy efficient than an equivalent house that complies with the Deemed-to-Satisfy (DtS) Provisions. This may be achieved using software. V2.6.2.2 allows the use of a broader range of software, in addition to House Energy Rating Software. To be acceptable, the software must be capable of determining the heating and cooling load of a house by modelling a number of characteristics of the house. In particular, the software must be capable of modelling the roof, walls and floor of the house, the glazing, ventilation, hours of occupation and the unique climatic and topographical conditions of the site.

Whether a particular software program or version can be used will depend upon whether it has been recognised as meeting the requirements of the National Construction Code (NCC) by the applicable State or Territory department responsible for building regulation or the building official.

QUESTION 2

How do I determine the climate zone that my house is located in?

Answer

The ABCB has produced a map of Australia, which shows the extent of each climate zone relative to local government boundaries and a number of significant cities and towns. The map appears under the definition of climate zone in Part 1.1 of Volume Two. Generally the climate zones follow local government boundaries.

A series of State and Territory climate zone maps are available on the ABCB website, www.abcb.gov.au. Further clarification of the climate zone of a particular location can be obtained from the applicable State or Territory department responsible for building regulation.

QUESTION 3

How do you determine how much insulation is required in the roof, walls and floor of a house?

Answer

Part 3.12.1 of Volume Two details the minimum Total R-Value that is required to be achieved by the roof, walls and floor of a house. Once the minimum Total R-Value for these elements is identified, the amount of insulation that is required to be added to these elements can be determined by subtracting the inherent R-Value of the construction materials from the minimum Total R-Value.

For example, if a roof is required to achieve a minimum Total R-Value of R5.1 and the Total R-Value of the roof and ceiling construction (e.g. roof tiles, airspace and ceiling plaster) is R0.41, the amount of insulation that is required to be added is R4.69 (R5.1 – R0.41).

There are a number of figures in Part 3.12.1 which provide the Total R-Value of common forms of roof, wall and floor construction.

When determining Total R-Values, the following points should be noted. Firstly, it is important to realise that an airspace within an element contributes to the Total R-Value of the construction. Therefore, if insulation is chosen which takes up all the airspace, the R-Value of the airspace needs to be subtracted from the Total R-Value of the construction. This is likely to occur when bulk insulation is chosen to be installed within a wall cavity.

Secondly the minimum Total R-Values given for roofs and floors in Part 3.12.1 also prescribe a "direction of heat flow" for that minimum Total R-Value. This means that when the insulation is selected, the insulation must achieve the required R-Value in the specified direction.

QUESTION 4

What if I am unable to install the required amount of insulation within my roof, walls or floor?

Answer

Part 3.12.1 contains concessions and options for each element that can be selected instead of simply complying with the required minimum Total R-Value. For example, in the warmer climate zones, a roof that is appropriately ventilated and has insulation located on the ceiling, gains a R0.5 reduction in the Total R-Value required to be achieved. As another example, in the milder climate zones, a house with cavity masonry external walls and masonry internal walls does not require as much insulation installed as would a masonry veneer wall or a framed wall.

Another option is to use the software approach to design a house. This allows the use of house energy rating software to demonstrate that a house achieves the required level of energy efficiency. The advantage of using software is that it provides much more flexibility in designing a house that is sufficiently energy efficient. For example, if a designer does not want to insulate the floor of a house, the software enables other options to be explored to compensate for this, such as smaller or better performing glazing.

QUESTION 5

Am I better off with a concrete slab-on-ground floor or a suspended floor?

Answer

The energy efficiency provisions allow the use of either a concrete slab-on-ground floor or a suspended floor. However, because of their different characteristics, the energy efficiency provisions treat each floor slightly differently. For example, a minimal level of insulation is required with all suspended floors, both concrete and timber while a floor coupled to the ground does not require insulation unless it contains an in-slab heating or cooling system, or is located in climate zone 8. The floor type also has a bearing on the wall and glazing provisions.

It should be noted that the V2.6.2.2 Verification Method or the energy rating DtS approach allows greater flexibility in designing complying energy efficient houses. This allows for other solutions to be explored, based on the desired floor type.

QUESTION 6

How do I ensure that I have purchased the right sort of insulation?

Answer

Insulation for use in houses is required to comply with AS/NZS 4859.1, a standard which prescribes the testing and labelling requirements for insulation. Insulation should be clearly labelled with a statement of compliance with AS/NZS 4859.1. It should also be labelled with the product R-Value, and provide two values if the R-Value varies depending on the direction of heat flow.

When selecting insulation it is important to ensure that the insulation itself achieves the minimum R-Value for the required direction of heat flow. The labelled R-Value may be for either the insulation material itself or for the Total R-Value of the installed insulation, which includes the R-Values of the construction materials e.g. roof and wall cladding and internal lining.

QUESTION 7

Why are the minimum Total R-Values for roofs and floors specified for a particular direction of heat flow?

Answer

Some insulation performs differently depending on the direction of heat flow through it. For this reason, the direction in which the minimum Total R-Value is to be achieved for roofs and floors is specified in Part 3.12.1.

A direction of heat flow into the building indicates that the house is located in a warm climate in which the house predominantly needs to resist the passage of heat into the house during normal hours of occupation. A direction of heat flow out from the house indicates that the house is located in a cool climate in which the house predominantly needs to resist the escape of heat during normal hours of occupation.

When purchasing insulation, it is important to check that it achieves the required R-Value for the specified direction of heat flow.

QUESTION 8

What is the easiest way of determining whether the glazing in a house is compliant?

Answer

The ABCB has developed a glazing calculator, which is available at the ABCB website, www.abcb.gov.au. The glazing calculator automates the calculations that are described in Part 3.12.2 of Volume Two. Users simply need to input the characteristics of the glazing (i.e. size, Total System U-Value, Total System SHGC and shading) and other details of the house into the glazing calculator and it will automatically apply the calculations in Part 3.12.2.

To further assist in determining compliance for the glazing in a house, the NCC provides conservative “worst case” performance values of the most common forms of glazing such as single clear glazing in aluminium or timber frames. The values are provided in the Explanatory Information following 3.12.2.1(b).

To assist practitioners to use the glazing calculator and achieve compliance, the ABCB has produced a Handbook ‘NCC Volume Two Energy Efficiency Provisions’, which can also be found on the ABCB website.

Compliance can also be determined using the software approach. A Youtube clip is also available.

QUESTION 9

The glazing calculator contains two columns headed “Element share of % of allowance used”. What does this mean in simple terms?

Answer

The glazing provisions in the NCC establish an allowance for conductance and an allowance for solar heat gain that cannot be exceeded by the glazing in the storey of a house. These two columns describe the contribution of each window or glazed door in relation to these allowances. The first column relates to the allowance for conductance while the second column relates to the allowance for solar heat gain.

In the glazing calculator, there are two numbers provided in each cell below these headings. The second number is the proportion of the allowance that has been achieved by all the glazing. A number over 100% indicates that the glazing collectively exceeds its allowance. A number less than 100% means that the glazing collectively is compliant. The first number is the contribution of the individual window or glazed door to the overall performance of the glazing in the storey. For example, if the first number was 50%, it would mean that that individual window or glazed door is contributing half of the overall performance of the glazing in the storey.

This is particularly useful information when the glazing exceeds its allowance because it identifies which individual window or glazed door is using up most of the allowance. In turn, this enables the user to determine which individual windows or glazed doors need to be addressed in order to bring the glazing back below the allowance.

QUESTION 10

How do I know if my windows and doors are adequately sealed?

Answer

External doors and openable windows serving habitable rooms must either be fitted with weather seals or comply with the air infiltration requirements in AS 2047 (refer Part 3.12.3.3). Weather seals may consist of compressible rubber strips, fibrous seals or the like. External swing doors must also have a seal on the bottom edge. Evidence of compliance with AS 2047 should be available from the supplier or manufacturer of the windows and glazed doors.

It is important to note that there are a number of instances in which a house does not need to comply with the above (refer Part 3.12.3). In particular, windows and doors need not be sealed if the house is located in the warmer/milder climates where the only air-conditioning system is an evaporative cooler (climate zones 1, 2, 3 and 5). This is because evaporative coolers rely on relieving the air to outside air in order to work effectively.

QUESTION 11

Do I have to add the ventilation opening area in the energy efficiency provisions to that required for health and amenity?

Answer

The energy efficiency air movement provisions (refer Part 3.12.4) and the ventilation provisions for health and amenity (refer Part 3.8.5) are separate standalone provisions both of which must be complied with. The energy efficiency air movement provisions are designed to facilitate the use of natural ventilation to cool houses. However, the ventilation provisions for health and amenity are designed to facilitate minimum liveable conditions for the building occupants. The one opening can serve both purposes.

For energy efficiency, Table 3.12.4.1 prescribes the minimum ventilation opening that is required to service each habitable room in a house. The amount of ventilation opening depends on the applicable climate zone and whether the room is served by an evaporative cooler or ceiling fan.

It is important to note that where a ventilation opening has been installed to comply with the health and amenity provisions, it may be counted towards compliance with Table 3.12.4.1.

QUESTION 12

When did the energy efficiency provisions come into force?

Answer

The energy efficiency provisions for houses were originally introduced into Volume Two on 1 January 2003. The stringency was increased to nominal 5 star on 1 May 2006 and further increased to 6 star on 1 May 2010.

However, it is important to note that the actual date that these provisions were enforced in each State and Territory has varied. The "History of Amendments" section at the rear of Volume Two provides general advice on the dates of adoption of the energy efficiency provisions in each State and Territory. It should also be noted that some States and Territories also introduced transitional periods during which compliance was not mandatory.

The applicable local building control authority or State or Territory department responsible for building regulation should be consulted for more precise details on the date at which the energy efficiency provisions for houses came into force.

QUESTION 13

How do I apply the energy efficiency provisions to an existing house?

Answer

The energy efficiency provisions only apply to new building work, that is, to new houses and to any new work in an existing house.

In terms of new work in existing houses, it should be noted that there may be circumstances in which a dispensation may be granted from complying with the energy efficiency provisions. This will depend on the applicable State or Territory regulations. However, there are a number of general principles that are likely to be consistently applied.

For example, if an alteration to an existing house does not involve removing the lining from the existing external walls, it may be argued that it is unreasonable to require insulation to be installed in these walls. Similarly, it may be considered unreasonable to require upgrading of the existing windows of a house when they are not affected by an alteration or addition being carried out.

For the purposes of calculating compliance of any new glazing, which forms part of an alteration or addition to an existing house, it is suggested that all glazing, both new and existing, be given the same performance values; accepting that only the new glazing will actually be installed to achieve these values. The reason for attributing the same performance values to the existing glazing is because compliance is calculated on a whole storey basis. In carrying out these calculations, it is likely to create a significant disadvantage if the actual performance values of the existing glazing are applied.

It should also be remembered that where it seems unreasonable to apply the Deemed-to-Satisfy Provisions to an alteration or addition to an existing building, the Verification Methods may be used to develop a more acceptable option. Development of a PerformanceSolution via other means may also be considered.

The applicable local building control authority or State or Territory department responsible for building regulation should be consulted for more specific advice about the application of the energy efficiency provisions to existing buildings.

QUESTION 14

Who can certify my energy efficiency design?

Answer

The Approval Authority is responsible for approving buildings. However, in many jurisdictions Approval Authorities may rely on certification from a specialist, such as an energy efficiency specialist, when making their determination.

Legislation in some States and Territories requires such specialists to be registered. It is recommended that the relevant State or Territory Department responsible for building regulation be contacted to obtain applicable requirements. Contact details for these Departments are available from the ABCB website, www.abcb.gov.au.